

Pipeline v2

Segment A: Calibration Run (20 docs, you grade)

First: Sample Selection

- Stratify the 20 deliberately:
 - 5 short (<200 pp), 10 medium (200–800), 5 long (800+)
 - Mix of EIS types: Draft, Final, Supplemental, ROD
 - Mix of bureaus (no more than 4 from any one)

M1: Easily Gathered (NUL-first, not LLM-first)

- Title
 - Primary: NUL
 - Fallback (if missing): Haiku, first page
 - Sanity check: non-empty, plausible length
 - Confidence: high (NUL) / medium (Haiku)
- Year (publication year of this volume — define explicitly)
 - Primary: NUL
 - Fallback: regex on first 3 pages
 - Sanity check: $1969 \leq \text{year} \leq \text{current year}$
 - Confidence: high (NUL) / medium (regex) / low (NUL and doc disagree → flag)
- EIS Type
 - Primary: regex on first page against closed vocabulary {Draft, Final, Supplemental, ROD}
 - Verifier: Sonnet on first 2 pages (NOT Haiku checking Haiku)
 - Confidence: high (regex + Sonnet agree) / medium (one source) / low (disagree)
- Contributor / Lead Agency (multi-valued — store as list)
 - Primary: NUL
 - Fallback: Sonnet, first 4 pages, output list
 - Flag if >2 agencies (likely joint-lead, usually correct but worth surfacing)

M2: Semantic Extraction

How we chunk:

1. PDF outline/bookmarks if present — most reliable
 2. Else regex for TOC + Haiku to label section ranges
 3. Else 50-page chunks with **2-page overlap**
- Where doc follows CEQ §1502, map to standard chapters: Purpose & Need, Alternatives, Affected Environment, Environmental Consequences, Mitigation, Consultation

Fields:

- Summary
 - Opus, section-mapped chunks in parallel, then reduce call
 - Schema: project_description, affected_community, alternatives_overview, environmental_impact, public_response
 - Each field cites pages
 - public_response carries explicit flag: based_on_main_doc_only (appendix comments out of scope)
- Alternatives Proposed
 - Operate on the Alternatives chapter identified structurally — NOT regex flag on the word "alternative"
 - Sonnet returns: list of alternatives, brief description each, page citations
- Themes
 - Sonnet, given chunk summaries (not raw text), assigns 1–3 themes + 2–5 subthemes from prepared taxonomy
 - Closed-set classification → confidence from rubric or self-reported with justification
- Location + Geometry
 - Sonnet on first 30 pages + any chapter titled "Project Area" / "Study Area"
 - Geocoder pass
 - For linear/multi-site projects: return list, not single point
 - Confidence: high (single named place, clean geocode) / low (multi-site, vague description)
- Key People / Groups — **split into three categories with different handling**
 - **Agency preparers**: deterministic extraction from Preparers chapter; high confidence; no stance attribution
 - **Cooperating agencies / tribal nations consulted**: deterministic list from Consultation chapter; no stance
 - **Public commenters with stance**: attempted only when main doc contains comment-response content; private individuals get last-name only or "private commenter"; stance + quote attempted only for clearly attributed positions
 - Merge step + verbatim quote checker against **exact** cited page (not ±2)

M2 Check: Critic (Sonnet)

- Receives: extracted field + citation + **the cited pages pulled in** (not just page numbers)
- Returns per field one of: PASS / PASS_WITH_NOTE / RE_EXTRACT / HUMAN_REVIEW
- Rubric per field — concrete checks, not 1–100:
 - Stance: (a) stance present on cited page? (b) quote verbatim? (c) named speaker clear?
 - Alternatives: each alternative discussed on cited page?
 - Themes: chosen themes supported by chunk summaries?

- Stakeholder stance for private individuals → **HUMAN_REVIEW** by default regardless of Critic verdict

M2.5: Grading Interface (Segment A only)

- Output a grading sheet per doc, one row per field
 - Columns: **field** | **extracted_value** | **source_pages** | **critic_verdict** | **model_confidence** | **your_grade** {correct, minor_issue, wrong, cant_tell} | **your_notes**
 - You grade all 20
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Segment B: Production Run

M-Cal: Calibration from your grades (the new step that uses Segment A output)

- Per field:
 - Accuracy = correct / total
 - Confidence-accuracy curve: threshold for **HUMAN_REVIEW** = lowest model confidence where observed grade accuracy $\geq$ target rate
 - Failure patterns from **your_notes** → added as explicit Critic checks
- Critic prompt updated with:
 - 2–3 few-shot examples per field drawn from your grades (one PASS, one HUMAN_REVIEW, one borderline)
 - Specific failure patterns surfaced as named checks
- Per-field accuracy targets locked in based on what's achievable:
 - Title, year, bureau, EIS type, theme → target $\geq$ 95%
 - Alternatives, summary, location → target $\geq$ 85%
 - Stakeholder stance (private individuals) → human review by default; no auto-accept target
- Human review queue gating thresholds: derived from data, not chosen

M1 → M2 → M2 Check: same pipeline as Segment A, now with calibrated Critic and confidence thresholds.

Re-calibration cadence: after every additional ~50 graded docs (you'll accumulate these from the review queue), rerun M-Cal. Drift is real and OCR-perfect doesn't protect against it — content patterns change.

M3: Deferred

- Appendices + standalone comment volumes (this is the big one — materially expands [public_response](#) and stakeholder coverage)
- Historical context / project status enrichment from external sources
- Cross-document linking: Supplemental ↔ original, ROD ↔ Final EIS
- Expanded eval set (your 20 → 100+ over time)